

Problem 1

Consider p_k the probability of getting \$20 at k dollars. We have a 0.5 chance to lose a dollar, 0.4 chance to gain a dollar, and a 0.1 chance to break even. Thus, $p_k = 0.5p_{k-1} + 0.4p_{k+1} + 0.1p_k$ thus

$$p_k = 5p_{k-1}/9 + 4p_{k+1}/9$$

$$p_{k+1} = 9p_k/4 - 5p_{k-1}/4$$

$$p_{k+1} - p_k = (5/4)(p_k - p_{k-1})$$

$$p_{k+1} - p_k = (5/4)^k(p_1 - p_0)$$

$$p_{k+1} - p_k = (5/4)^k(p_1)$$

Consider $\sum_{i=0}^k (p_k - p_{k-1})$

$\sum_{i=1}^k (p_i - p_{i-1}) = (p_k - p_{k-1}) + (p_{k-1} - p_{k-2}) + \cdots + (p_1 - p_0) = p_k - p_0 = p_k$ also from above we

know $p_i - p_{i-1} = (5/4)^i(p_1)$

$$p_k = \sum_{i=1}^k (p_i - p_{i-1}) = (p_1) \sum_{i=1}^k (5/4)^{i-1} = (p_1) \sum_{i=0}^k (5/4)^i = (p_1) \frac{1 - (5/4)^{k+1}}{1 - (5/4)}$$

$$p_k = (p_1) \frac{1 - (5/4)^{k+1}}{1 - (5/4)}$$

$$p_{20} = 1 = (p_1) \frac{1 - (5/4)^{21}}{1 - (5/4)}$$

$$p_1 = \frac{1 - (5/4)}{1 - (5/4)^{21}}$$

$$p_{10} = \frac{1 - (5/4)}{1 - (5/4)^{21}} \frac{1 - (5/4)^{11}}{1 - (5/4)} = \frac{1 - (5/4)^{11}}{1 - (5/4)^{21}}$$

Problem 2

$$G_1(s) = s/3G_2(s) + s/3G_1(s) + s/3G_0(s)$$

$$G_1(s) = s/3G_2(s) + s/3G_1(s) + s/3$$

$$\text{since } G_0(s) = E[s^{X_0}] = E[s^0] = 1$$

$$G_1(s) - s/3G_1(s) = s/3(G_2(s) + 1)$$

$$G_1(s)(1 - s/3) = s/3(G_2(s) + 1)$$

Given we know that for 1,3 we are both 1 away from the end points and have the same probability of going up, down, or breaking even, we know by symmetry that $X_1 = X_3$ thus $G_1(s) = G_3(s)$

$$\text{Thus } G_2(s) = s/3G_3(s) + s/3G_1(s) + s/3G_2(s)$$

$$G_2(s) = 2s/3G_1(s) + s/3G_2(s)$$

$$G_2(s) = 2s/(3 - s)G_1(s)$$

Thus plugging in G_2 we get that

$$G_3(s) = G_1(s) = \frac{3s - s^2}{9 - 6s - s^2}$$

$$G_2(s) = \frac{2s^2}{9 - 6s - s^2}$$

Problem 3

$$E[X|\text{starting at } HH] = (1/2)(1 + E[X|\text{starting at } HH]) + (1/2)$$

$$E[X|\text{starting at } HH] = 2$$

$$E[X] = E[X | T]\mathbb{P}(T) + E[X | HT]\mathbb{P}(HT) + E[X | HHT]\mathbb{P}(HHT) + E[X | HHH]\mathbb{P}(HHH)$$

$$E[X] = 1/2E[X + 1] + 1/4E[X + 2] + E[3]1/8 + (3 + E[X|\text{starting at } HH])1/8$$

$$E[X] = 1/2E[X] + 1/2 + 1/4E[X] + 1/2 + 3/8 + 3/8 + 2/8$$

$$1/4E[X] = 2$$

$$E[X] = 8$$

Problem 4

$$G_n(s) = spG_{n-1} + s(1-p)G_n$$

$$G_n(s) = \left(\frac{sp}{1-(p-1)s} \right) G_{n-1}$$

$$G_n(s) = \left(\frac{sp}{1-(p-1)s} \right)^n G_0$$

$$G_0(s) = 1 \text{ since there is one way to flip zero heads}$$

$$G_n(s) = \left(\frac{sp}{1-(p-1)s} \right)^n$$

$$G_n(s) = s^n p^n \left(\frac{1}{1-(p-1)s} \right)^n$$

$$G_n(s) = s^n p^n \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{k} (-(1-p)s)^k$$

$$G_n(s) = s^n p^n \sum_{k=0}^{\infty} \binom{n+k-1}{k} (1-p)^k s^k$$

$$G_n(s) = \sum_{k=0}^{\infty} \binom{n+k-1}{k} p^n (1-p)^k s^{k+n}$$

$$\text{Let } k = k+n$$

$$G_n(s) = \sum_{k=n}^{\infty} \binom{k-1}{n-1} p^n (1-p)^{k-n} s^k$$

$$\text{Thus } \mathbb{P}(X_n = k) = \binom{k-1}{n-1} p^n (1-p)^{k-n} \text{ for } k \geq n \text{ and } 0 \text{ for } k < n$$